SCALING SMARTLY: AWS AUTO SCALING GROUP (ASG) PROJECT DOCUMENTATION

⚙️ 🚀 🔥

BY KHUSHI NANDWANI

OVERVIEW 🌍

Auto Scaling Groups (ASG) in AWS allow you to automatically scale your EC2 instances based on demand, ensuring high availability and cost efficiency. This guide will walk you through setting up an ASG using a launch template and configuring scaling policies.

PREREQUISITES 🛠

Before proceeding, ensure you have the following:

✅ An active AWS account.

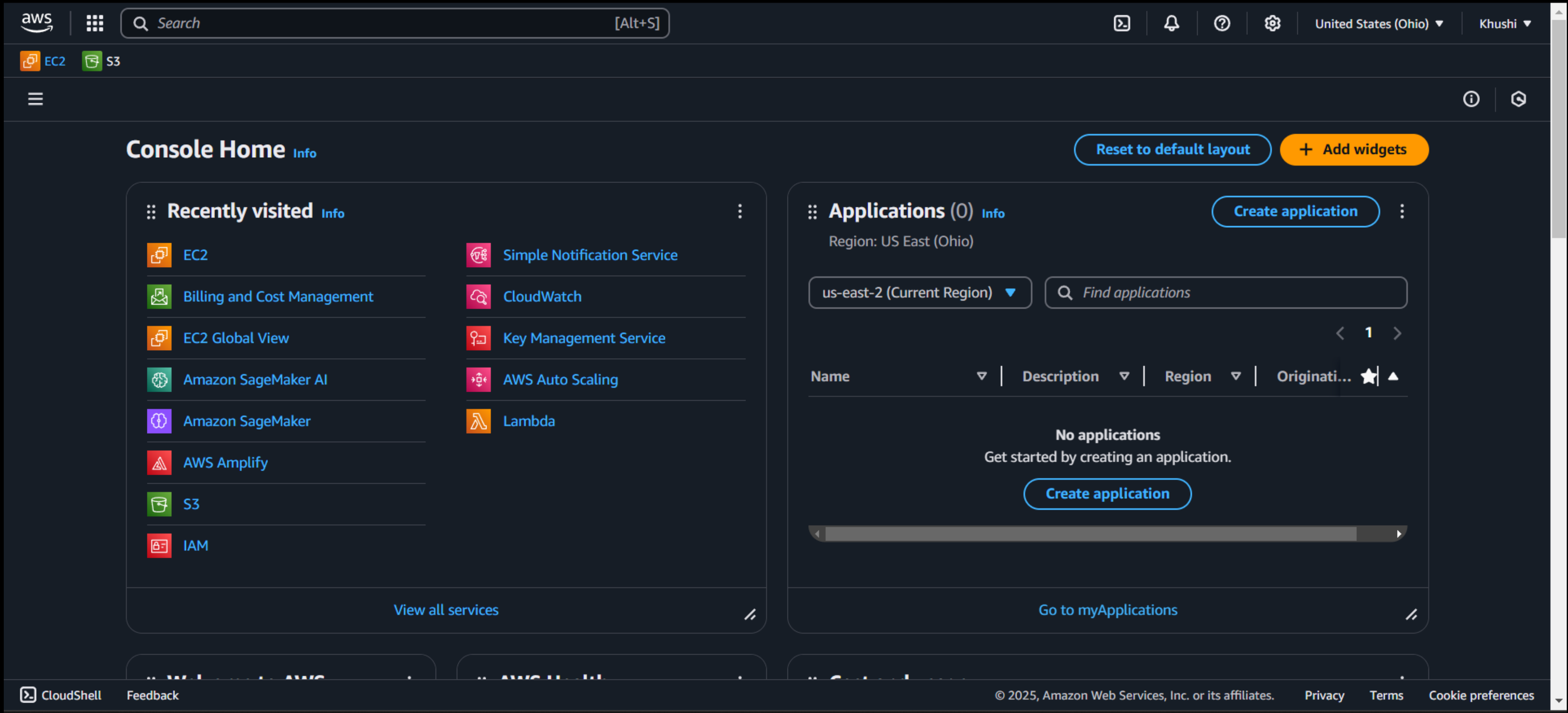
✅ Basic knowledge of AWS EC2 and Auto Scaling Groups.

✅ An IAM user with necessary permissions to create EC2 instances and ASGs.

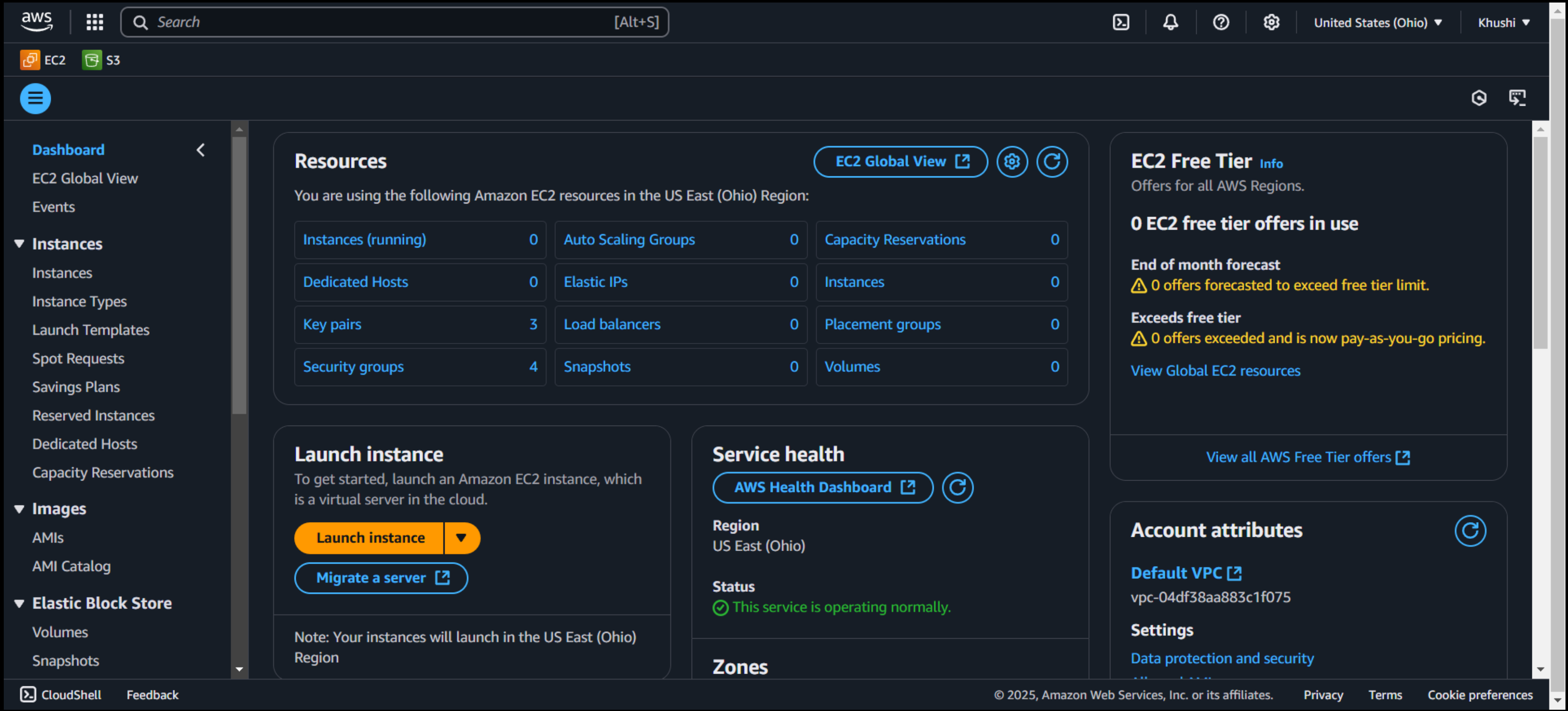
✅ A configured key pair for SSH access (if needed).

I. ACCESSING AWS MANAGEMENT CONSOLE 💻

1. Go to the AWS Management Console.

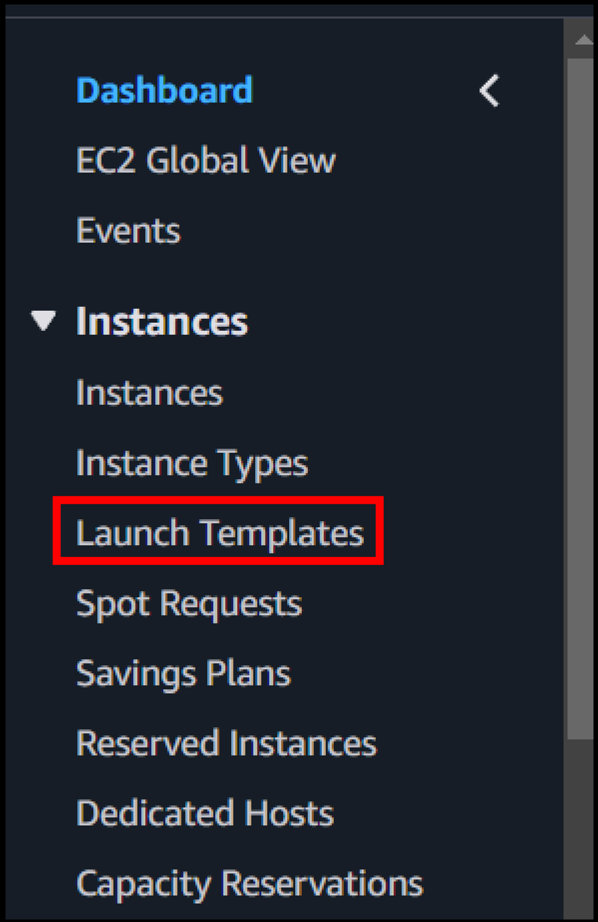


2. Search for EC2 and navigate to the EC2 dashboard.

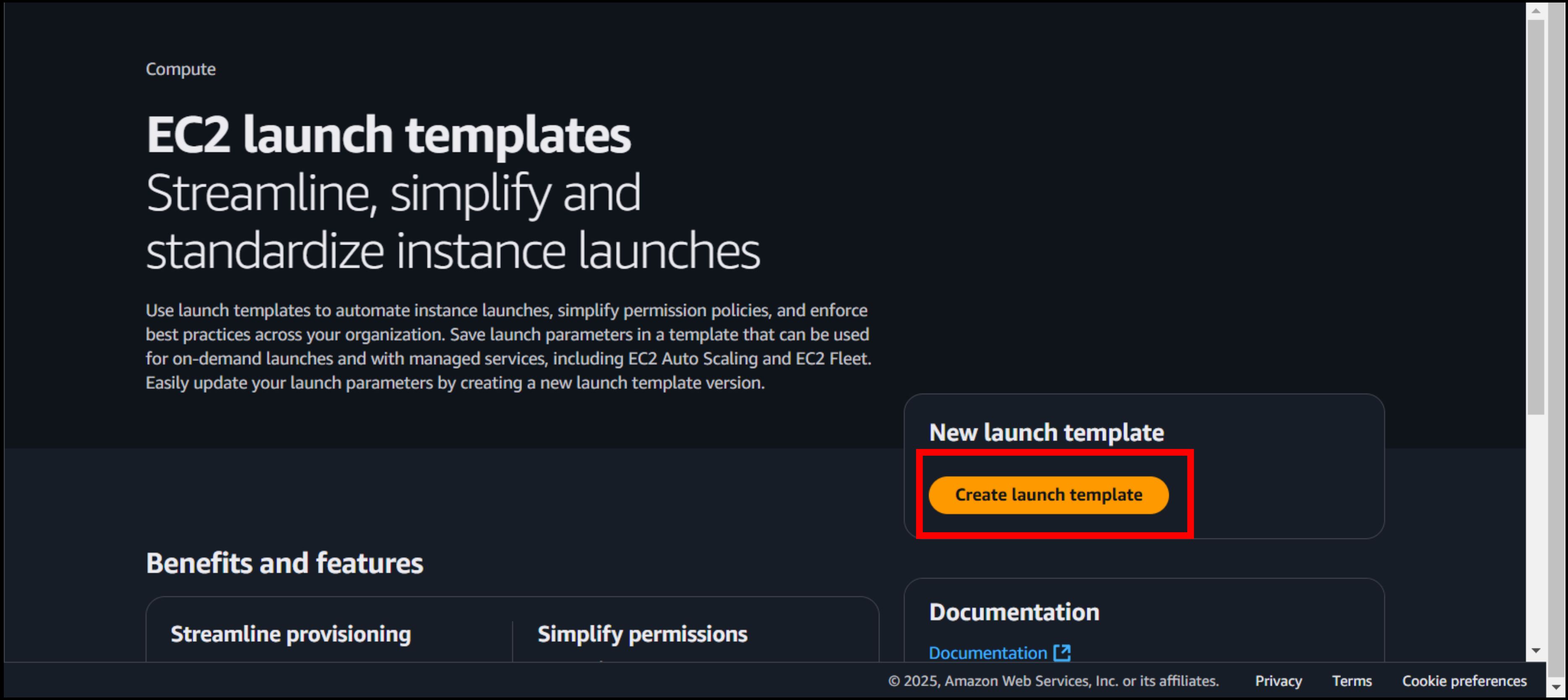


II. CREATING A LAUNCH TEMPLATE 🌐

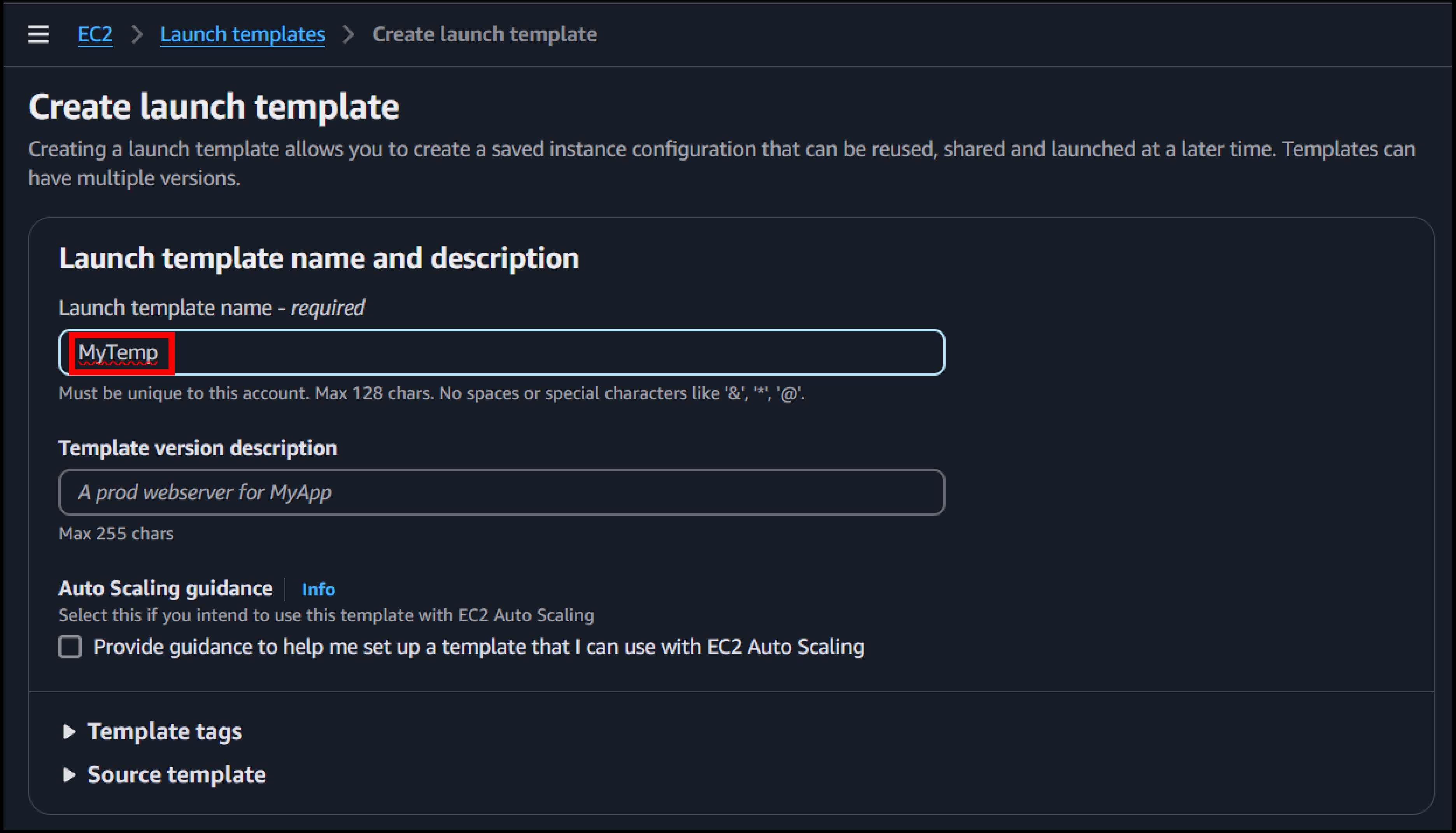
1. On the left-hand panel, click on "Launch Templates".



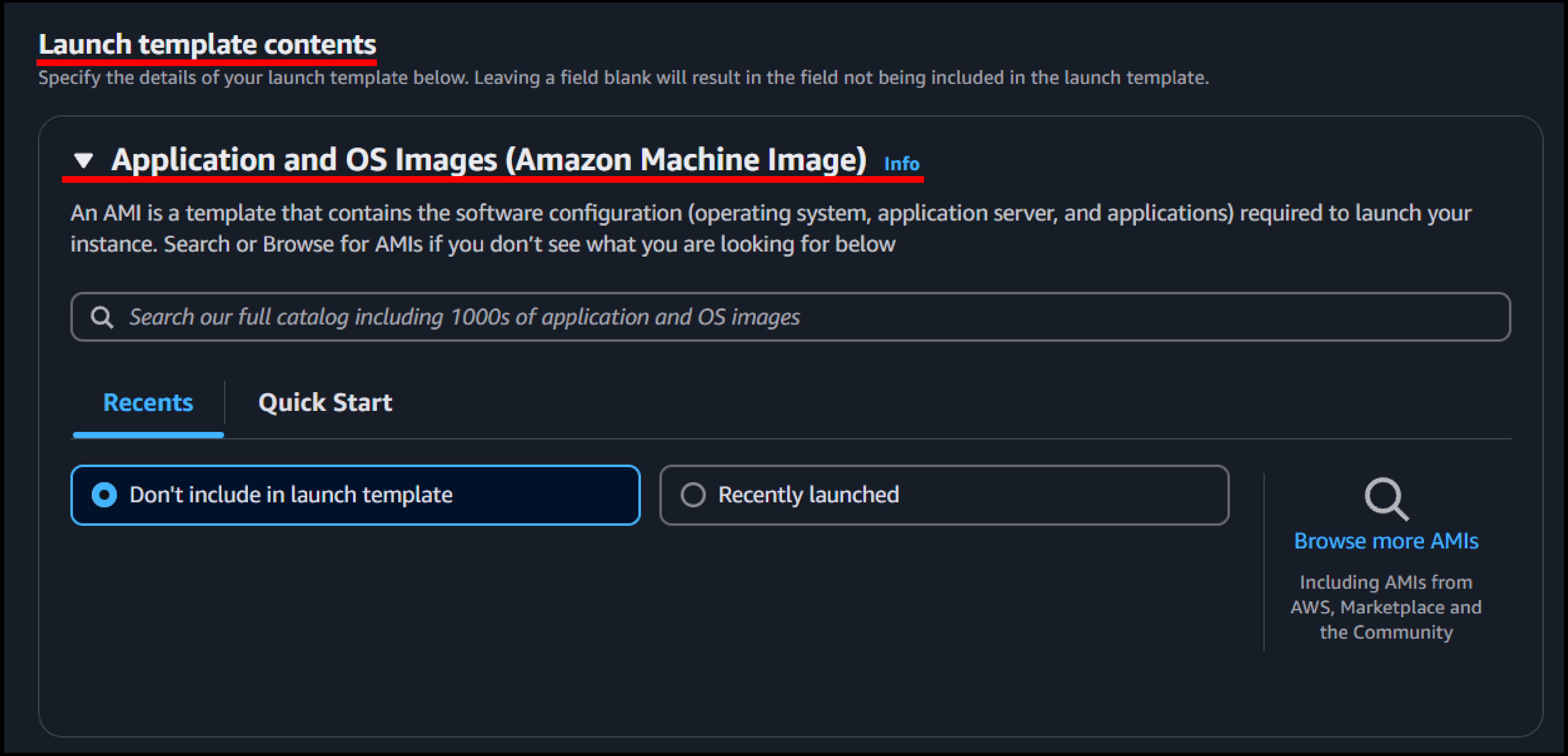
2. Click on "Create launch template".



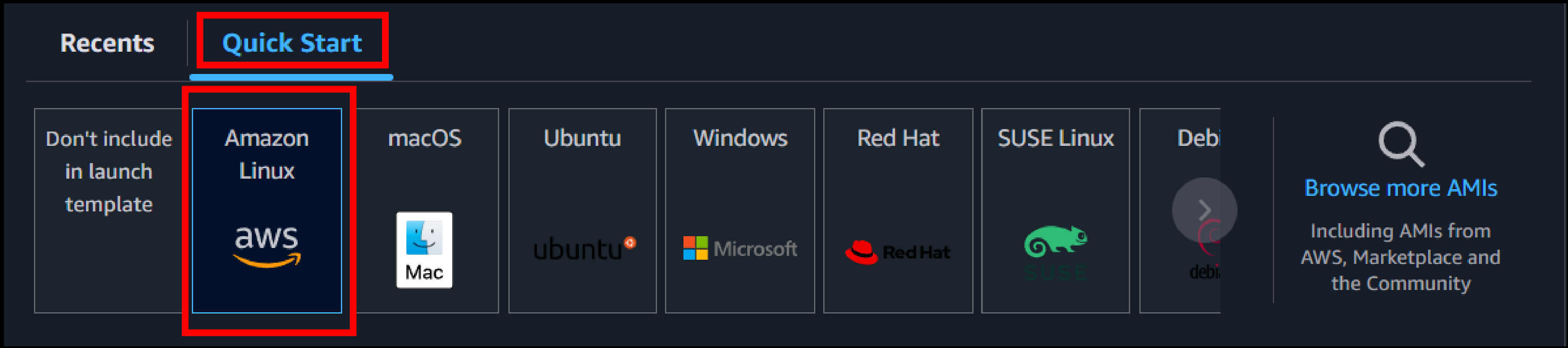
3. Provide a name for the template (e.g., MyTemp).



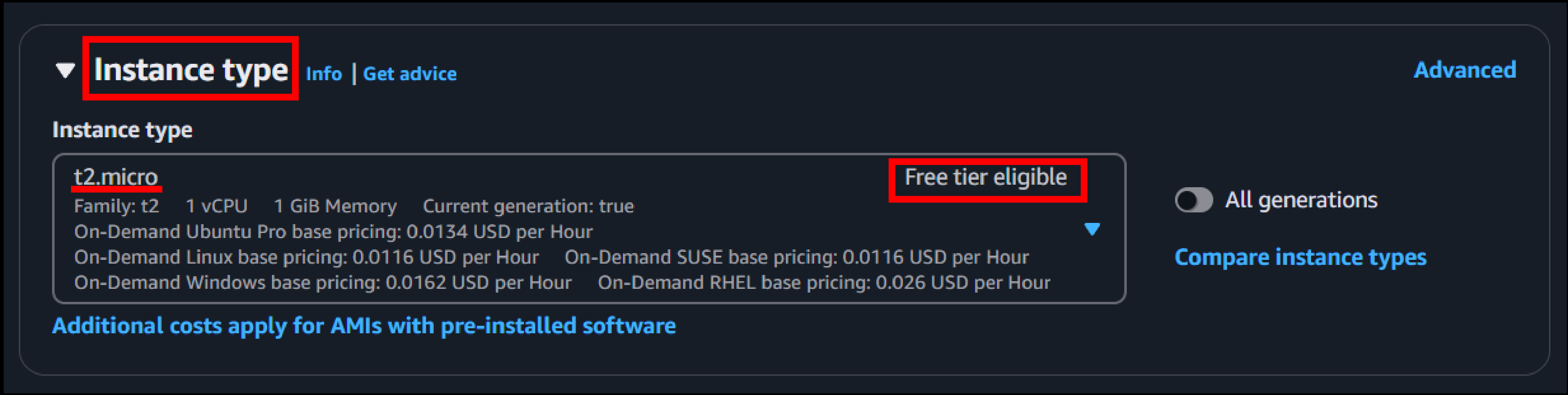
4. Scroll down to the AMI (Amazon Machine Image) section in Launch template contents.



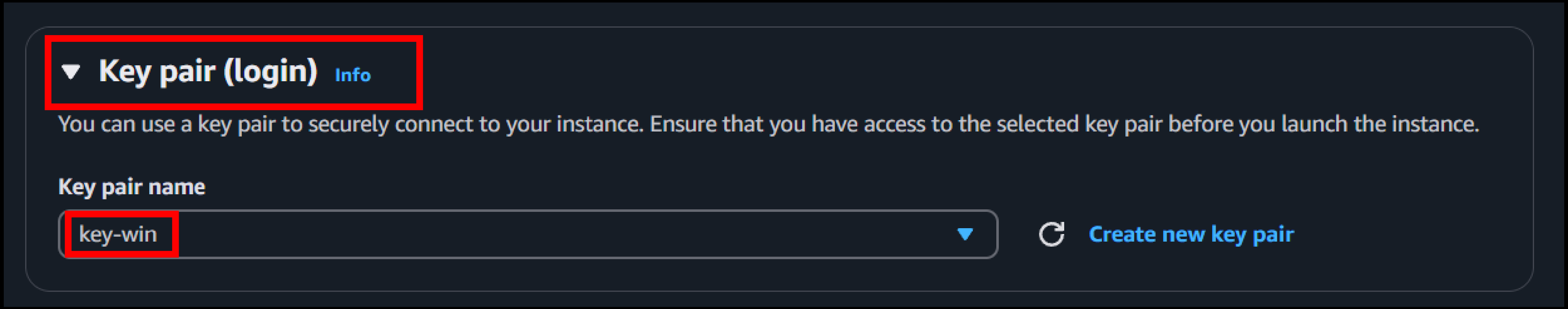
5. Click on "Quick Start" and choose Amazon Linux.



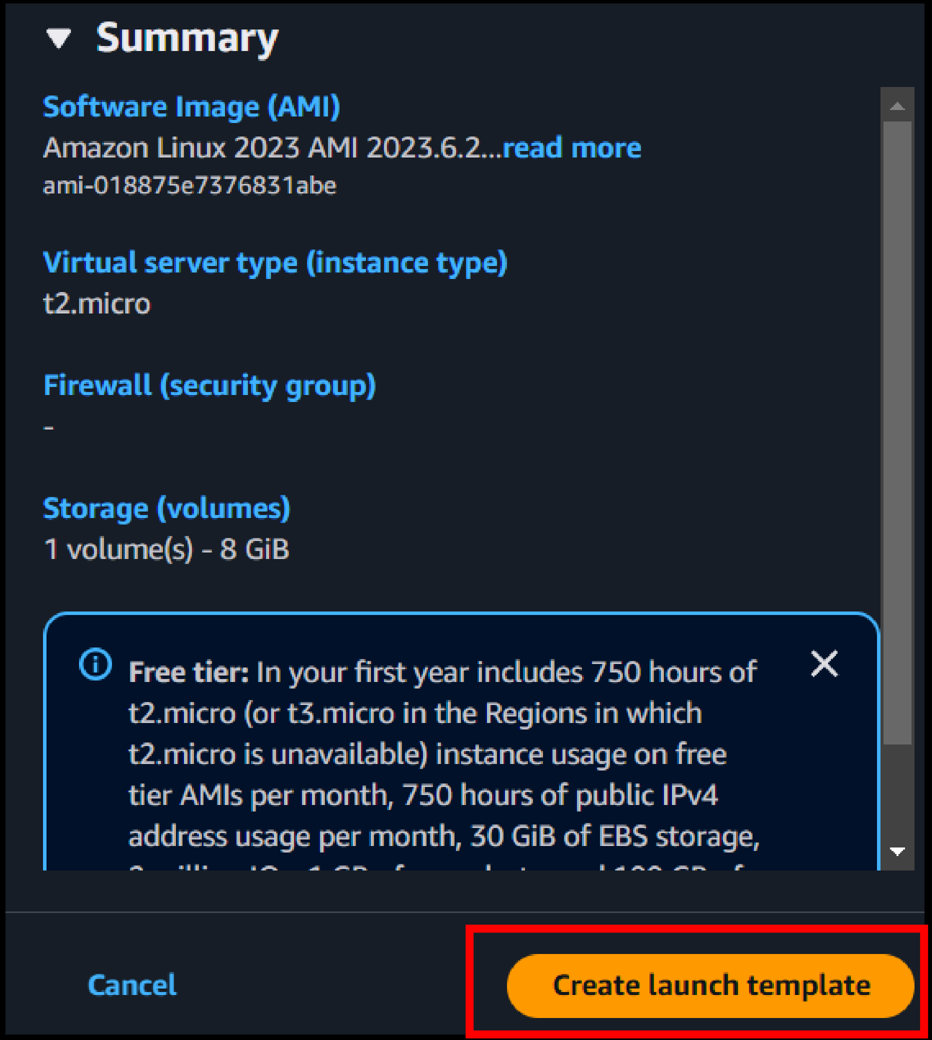
6. Select the instance type as t2.micro (free tier eligible).



7. Include a Key Pair for SSH access.

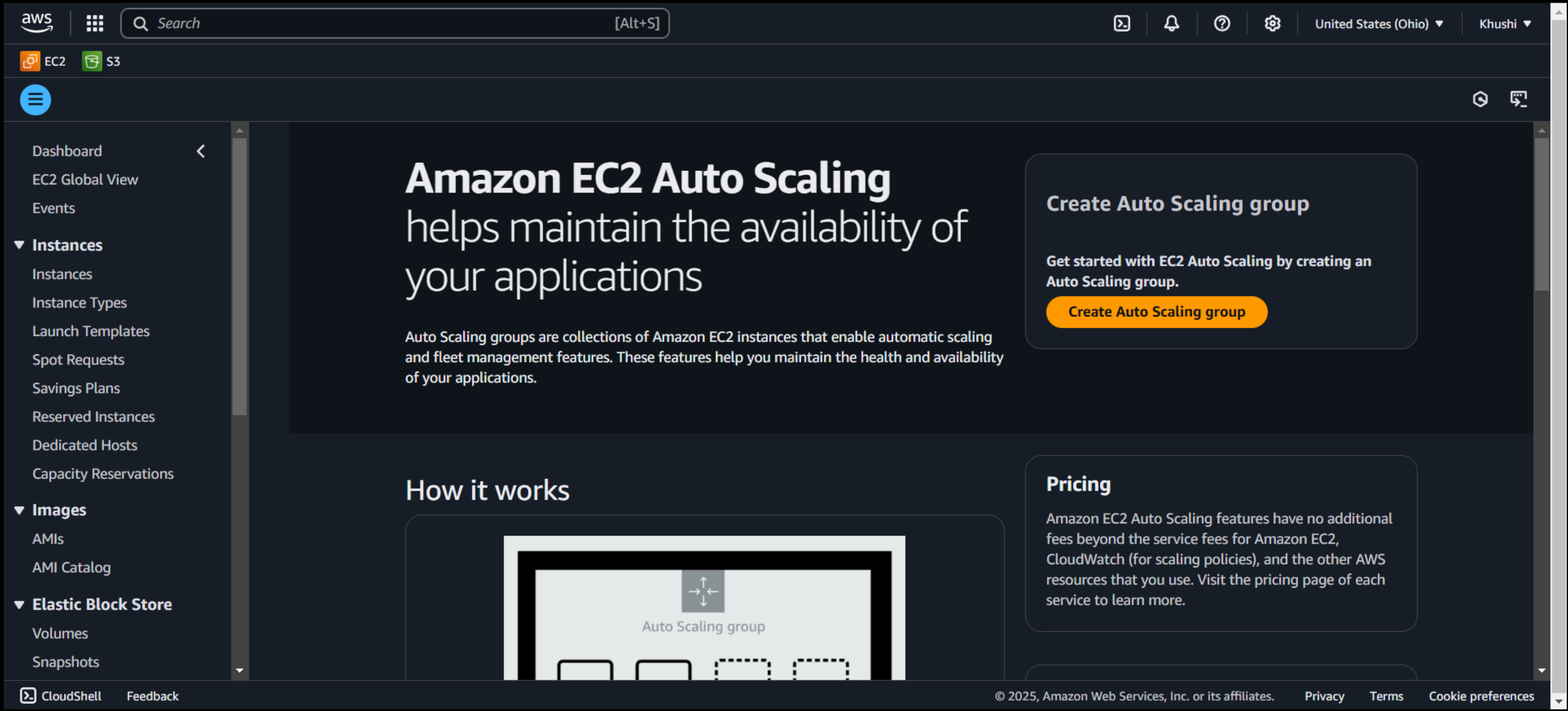


8. Keep the rest of the setting and same and In Summary click on "Create launch template".

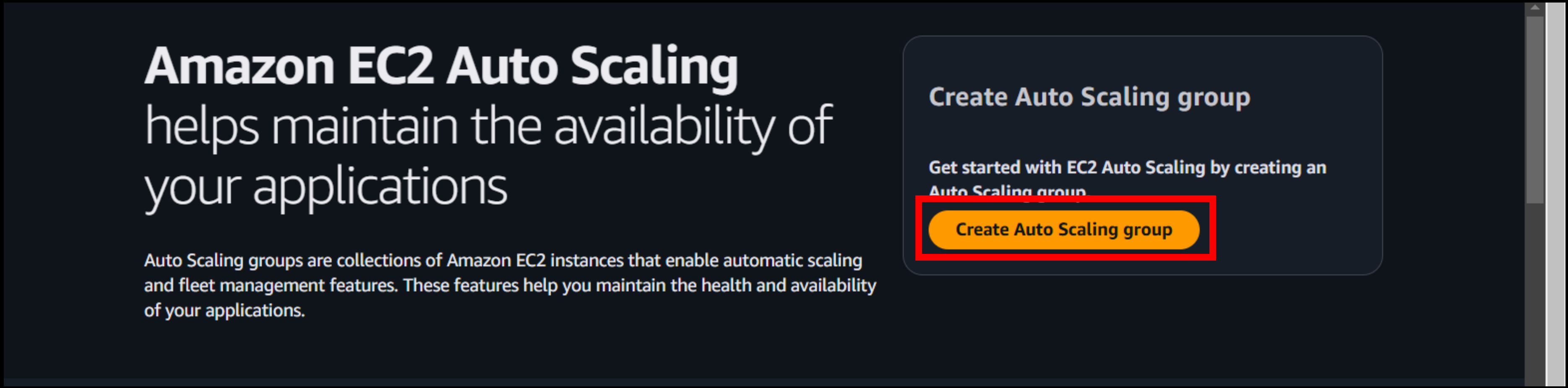


III. SETTING UP AUTO SCALING GROUP (ASG) ⚖️

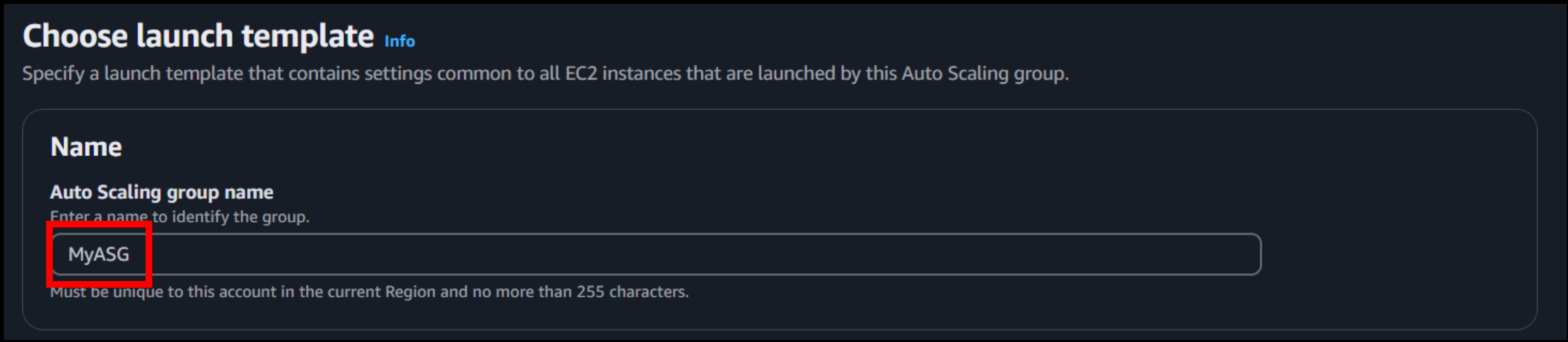
1. Navigate to ASG (Auto Scaling Groups) in the AWS console.



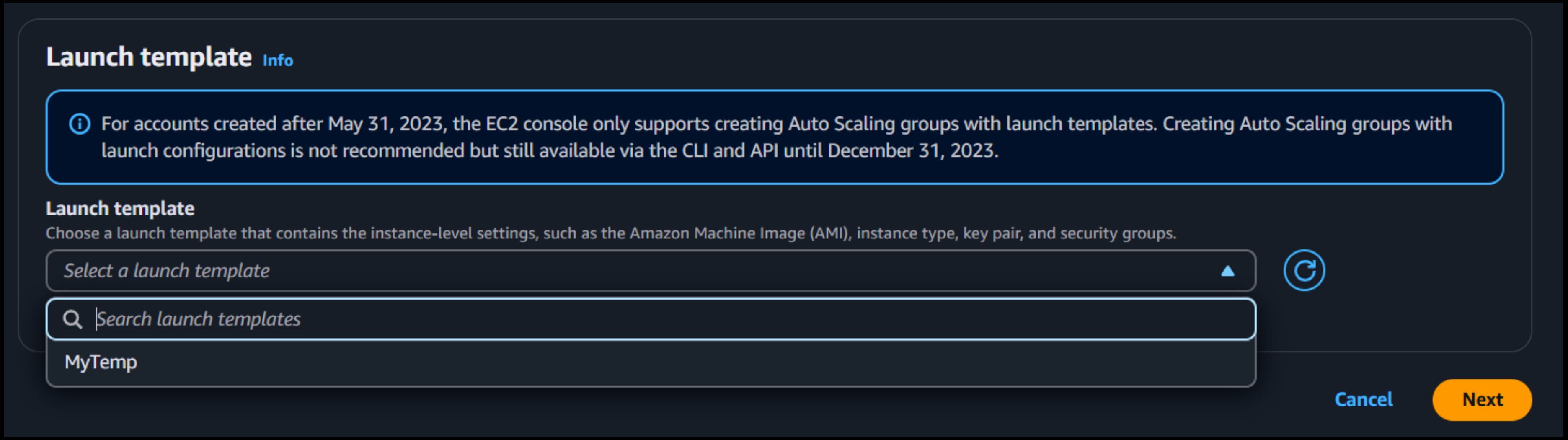
2. Click on "Create Auto Scaling Group".

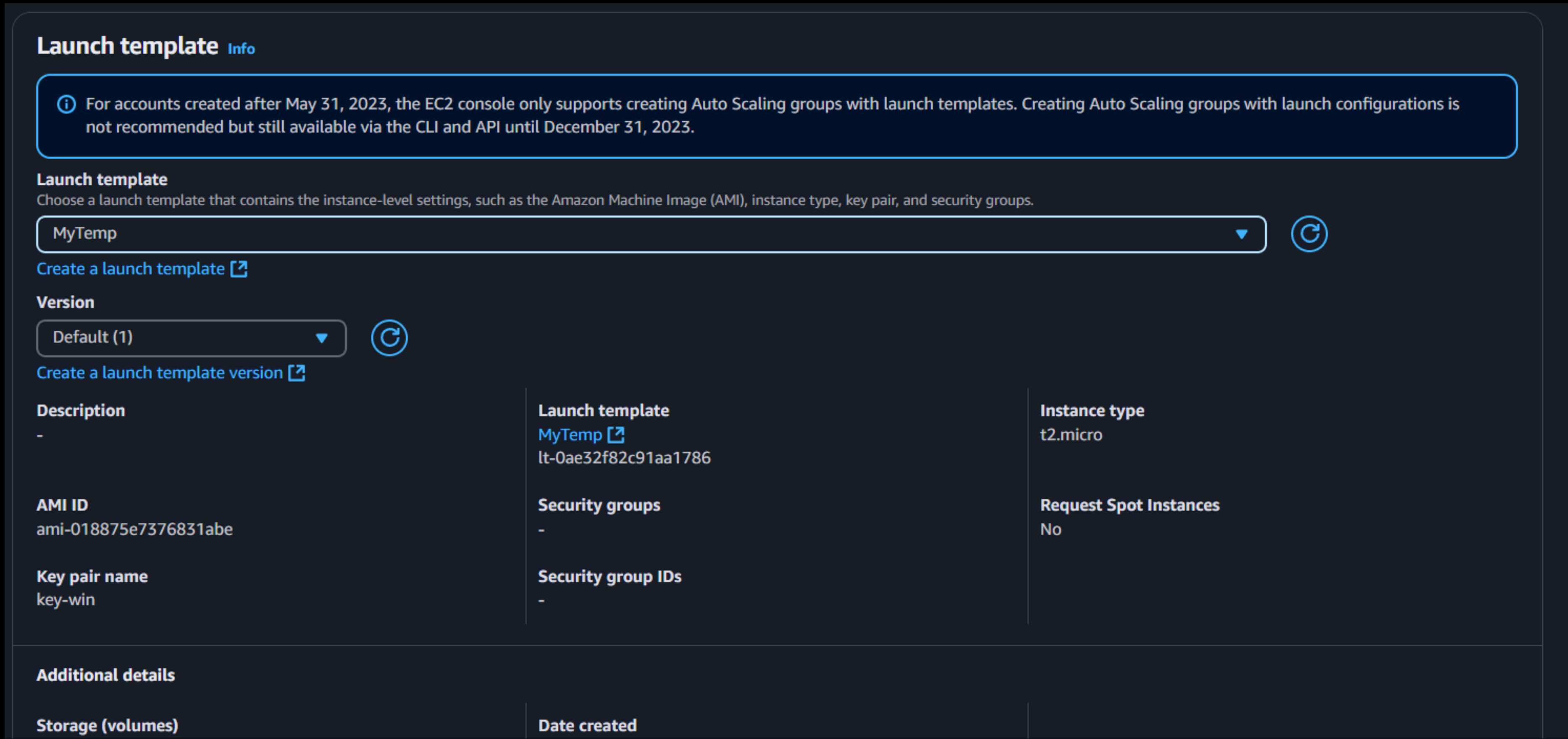


3. Provide a name for your ASG.

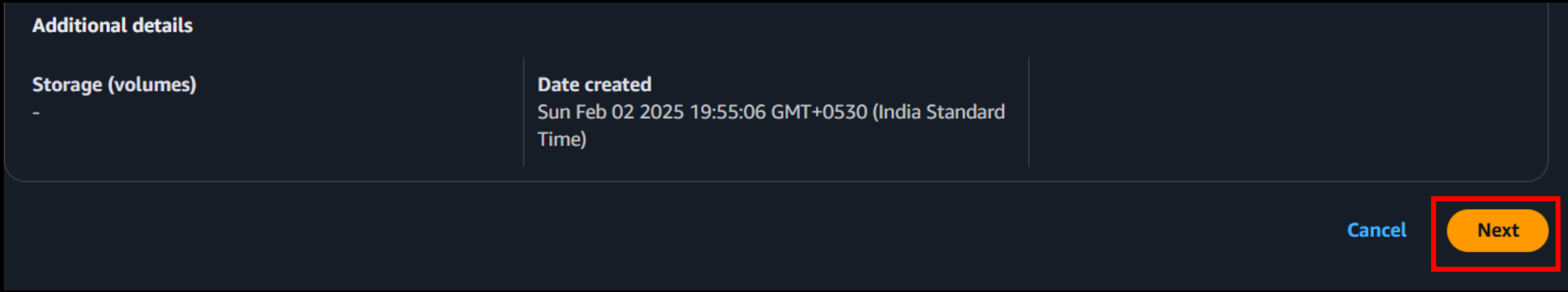


4. In the launch template selection, choose the previously created launch template.

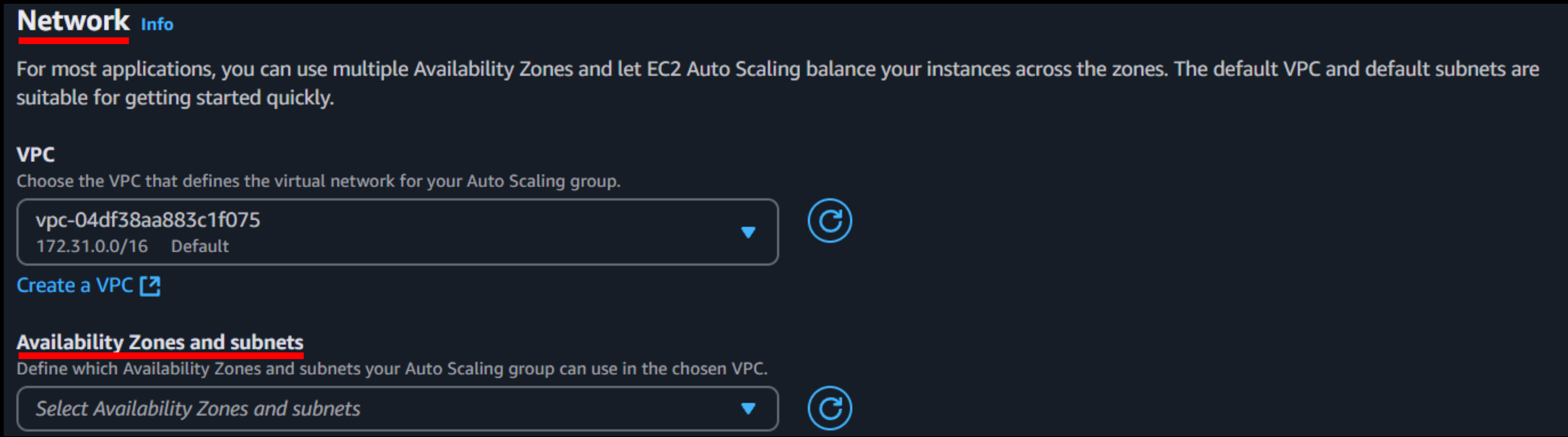


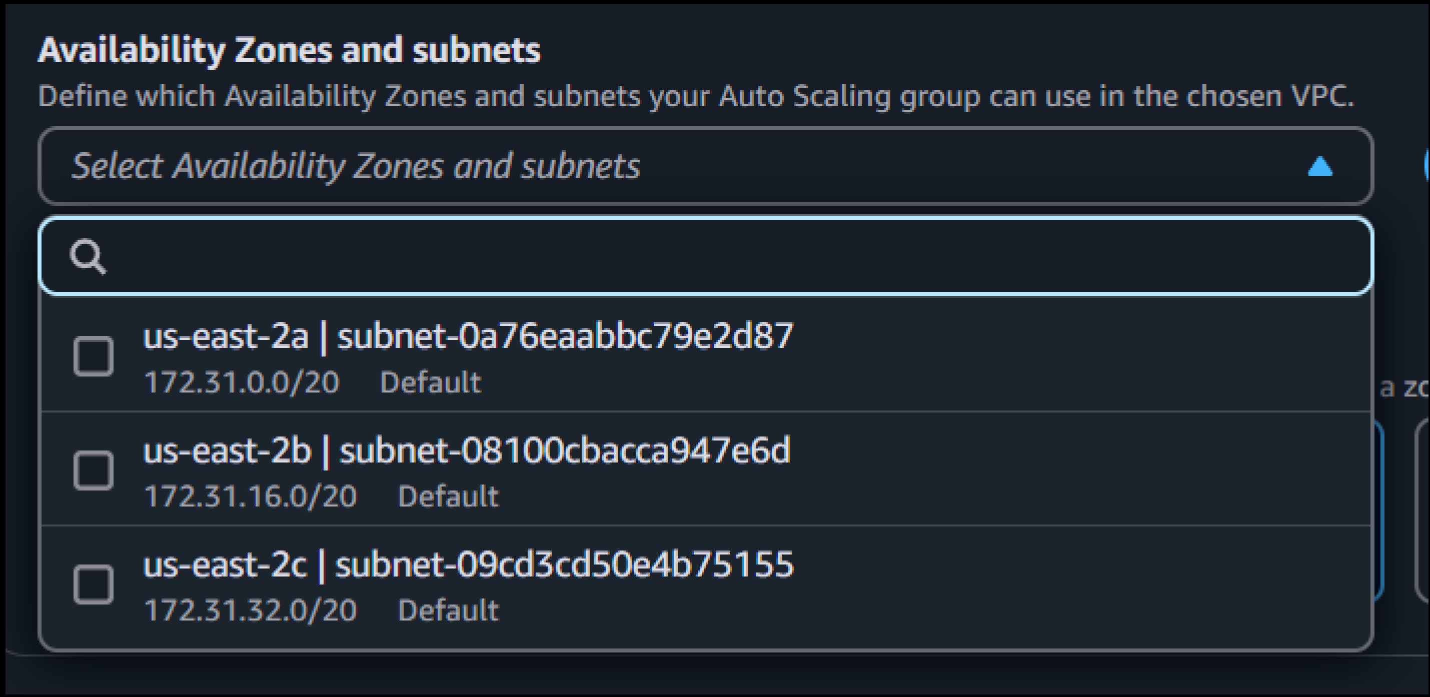

Launch template Info
For accounts created after May 31, 2023, the EC2 console only supports creating Auto Scaling groups with launch templates. Creating Auto Scaling groups with launch configurations is not recommended but still available via the CLI and API until December 31, 2023.
Launch template
Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.
MyTemp
Create a launch template
Version
Default (1)
Create a launch template version
Description
-
Launch template
MyTemp
lt-0ae32f82c91aa1786
Instance type
t2.micro
AMI ID
ami-018875e7376831abe
Security groups
-
Request Spot Instances
No
Key pair name
key-win
Security group IDs
-
Additional details
Storage (volumes)
Date created

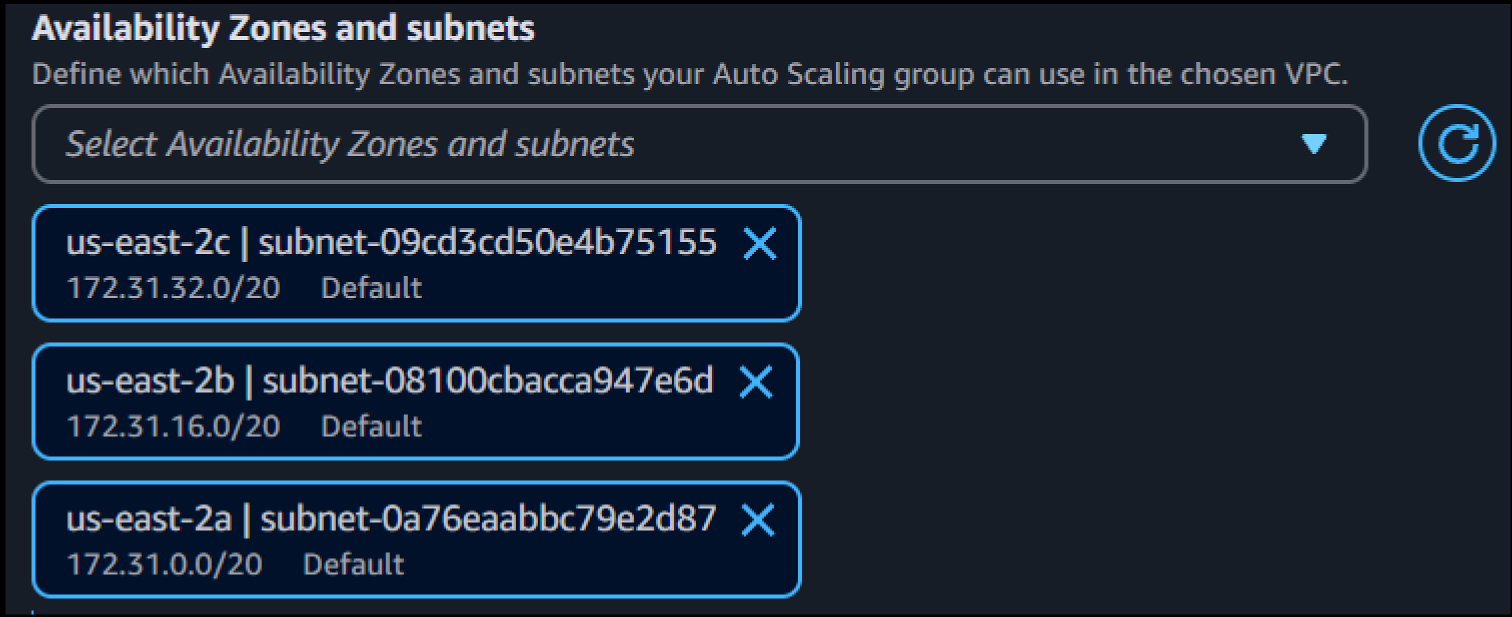
5. Scroll down and click on "Next".



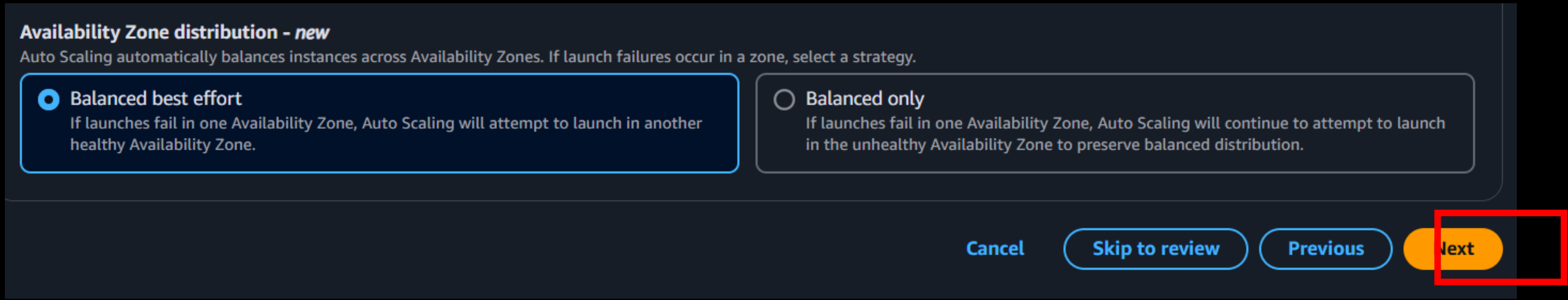
6. In the Network settings, select the Availability Zones (AZs) and click "Next".



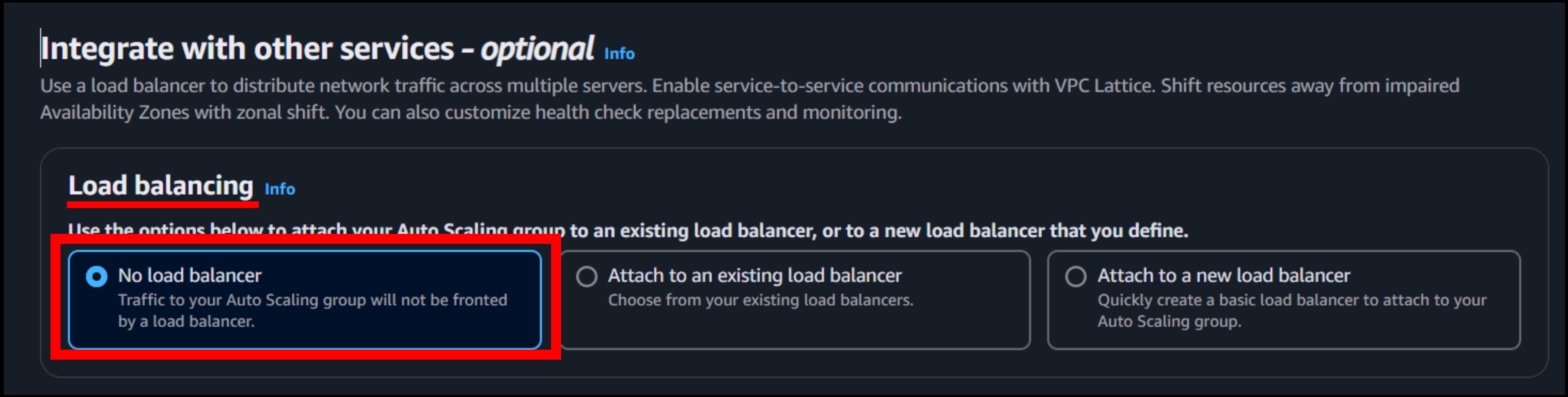

Availability Zones and subnets
Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.
Select Availability Zones and subnets
us-east-2a | subnet-0a76eaabbc79e2d87
172.31.0.0/20 Default
us-east-2b | subnet-08100cbacca947e6d
172.31.16.0/20 Default
us-east-2c | subnet-09cd3cd50e4b75155
172.31.32.0/20 Default


Availability Zones and subnets
Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.
Select Availability Zones and subnets
us-east-2c | subnet-09cd3cd50e4b75155
172.31.32.0/20 Default
us-east-2b | subnet-08100cbacca947e6d
172.31.16.0/20 Default
us-east-2a | subnet-0a76eaabbc79e2d87
172.31.0.0/20 Default

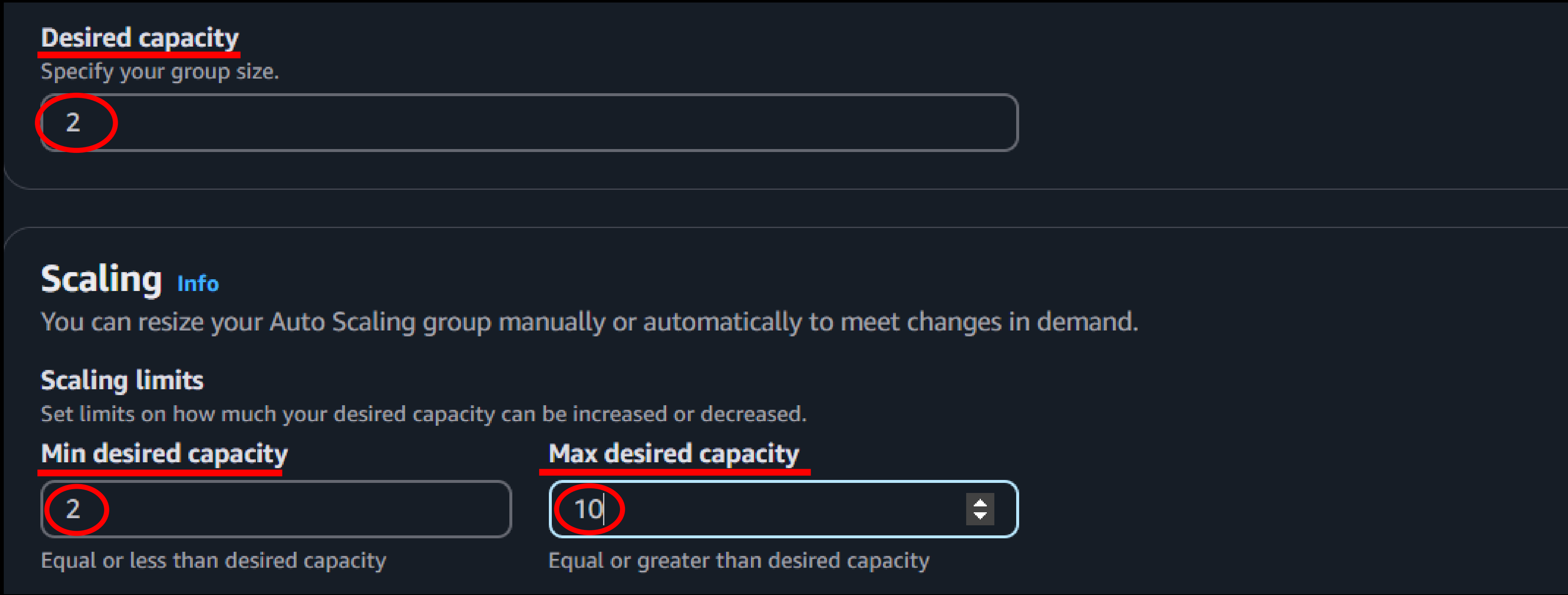
7. Click “Next”


Availability Zone distribution - new
Auto Scaling automatically balances instances across Availability Zones. If launch failures occur in a zone, select a strategy.
Balanced best effort
If launches fail in one Availability Zone, Auto Scaling will attempt to launch in another healthy Availability Zone.
Balanced only
If launches fail in one Availability Zone, Auto Scaling will continue to attempt to launch in the unhealthy Availability Zone to preserve balanced distribution.
Cancel
Skip to review
Previous

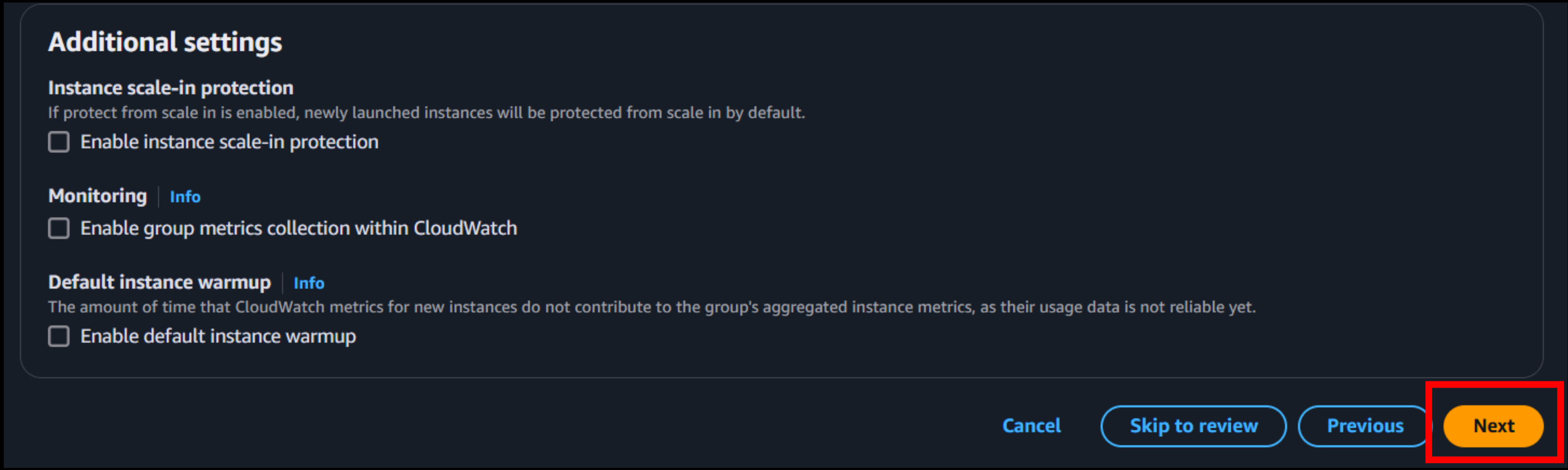
8. In the Load Balancers section, keep the default setting (No Load Balancer) and click "Next".



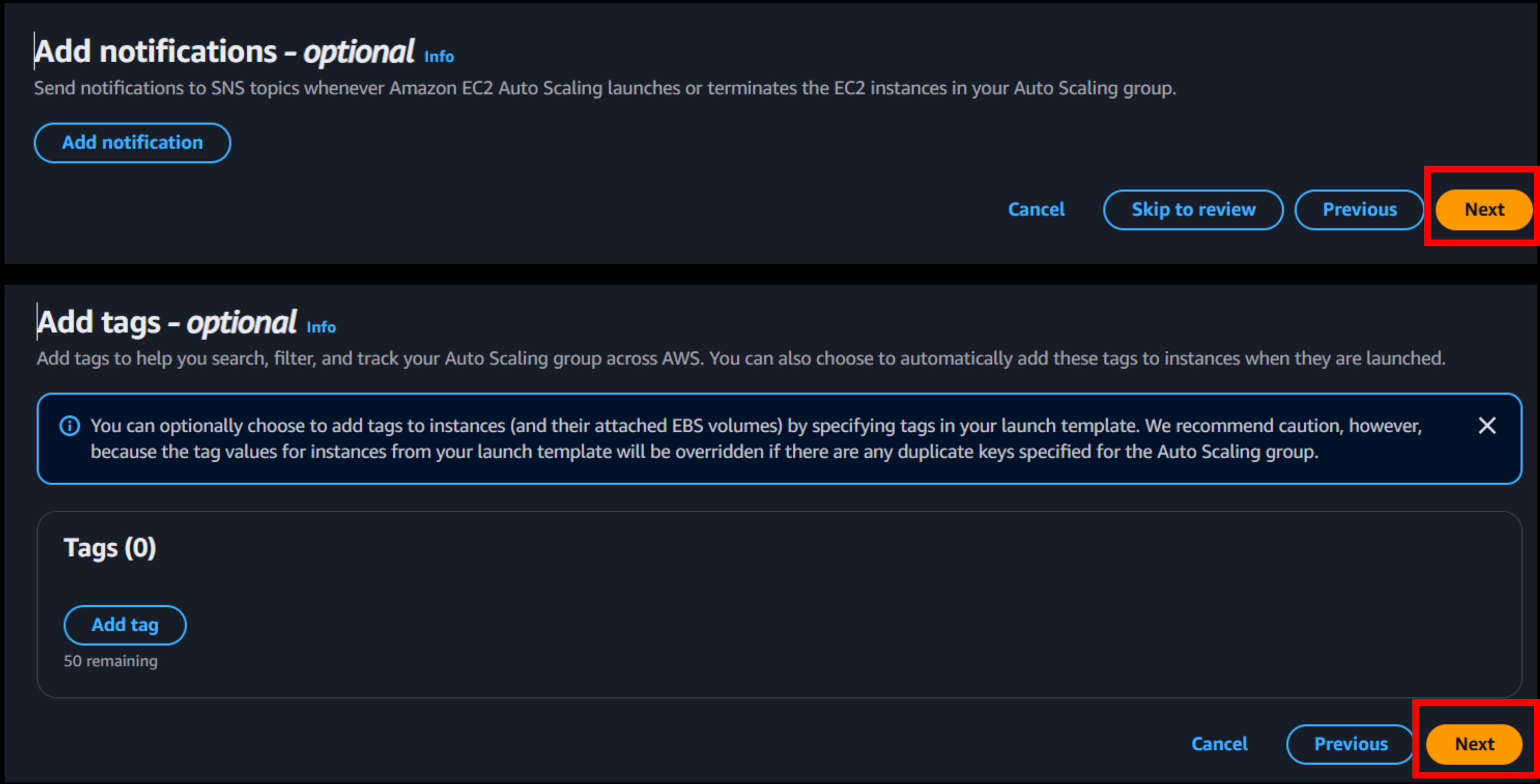
9. In the "Configure Group Size and Scaling" section, set the **desired capacity** as **2**, **Min desired capacity** as **2** & **Max desired capacity** as **10**.



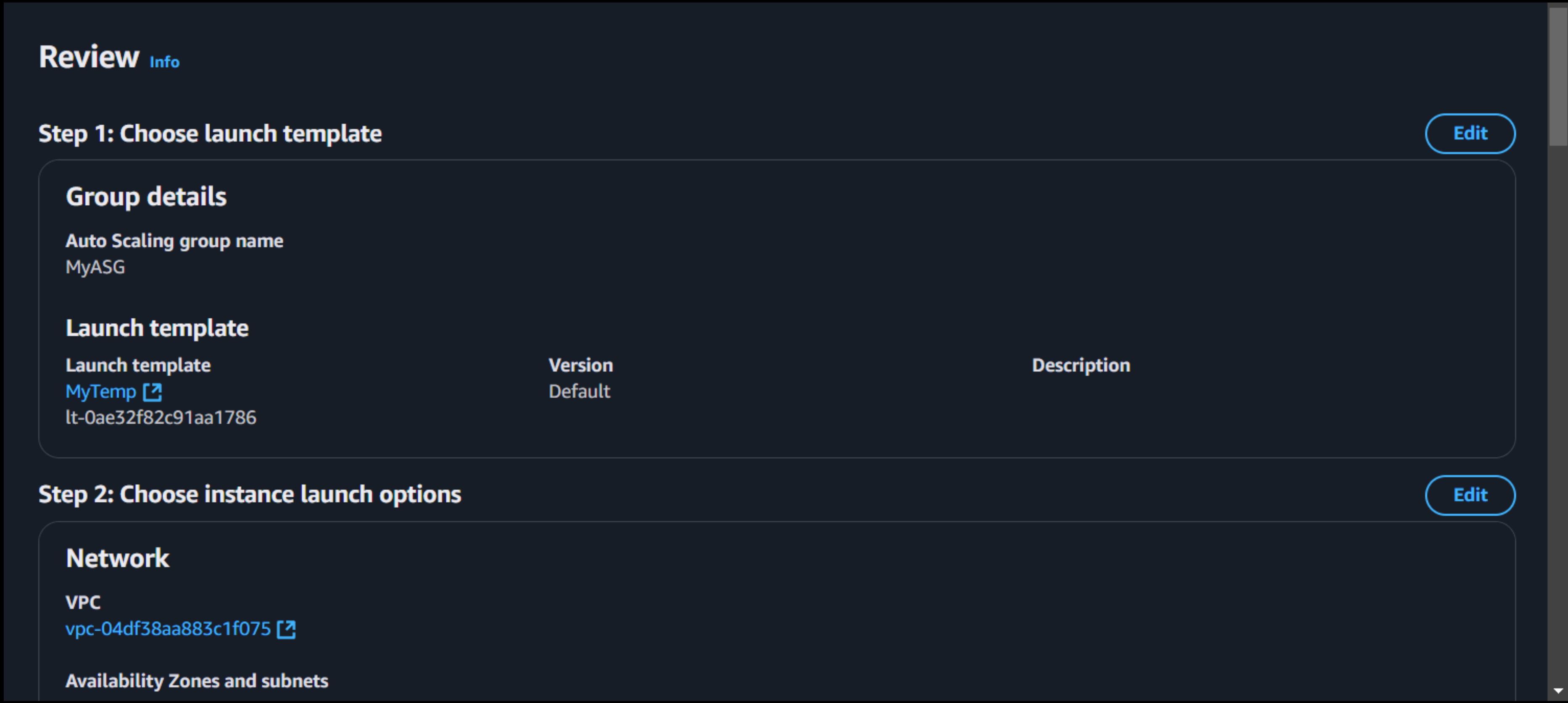
10. Keep everything else same, scroll down and click "Next".



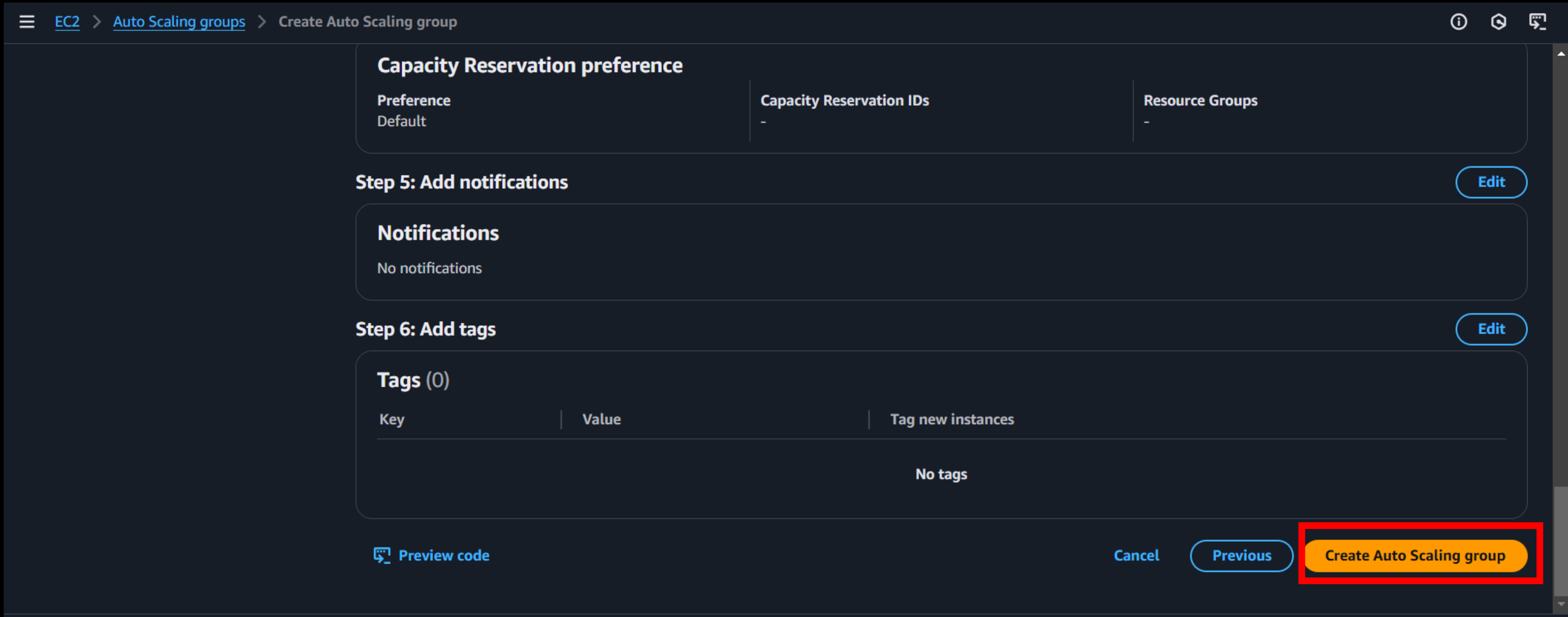
11. Click "Next" on the 5th and 6th steps as they are optional.



12. Review the configurations in the7th step.

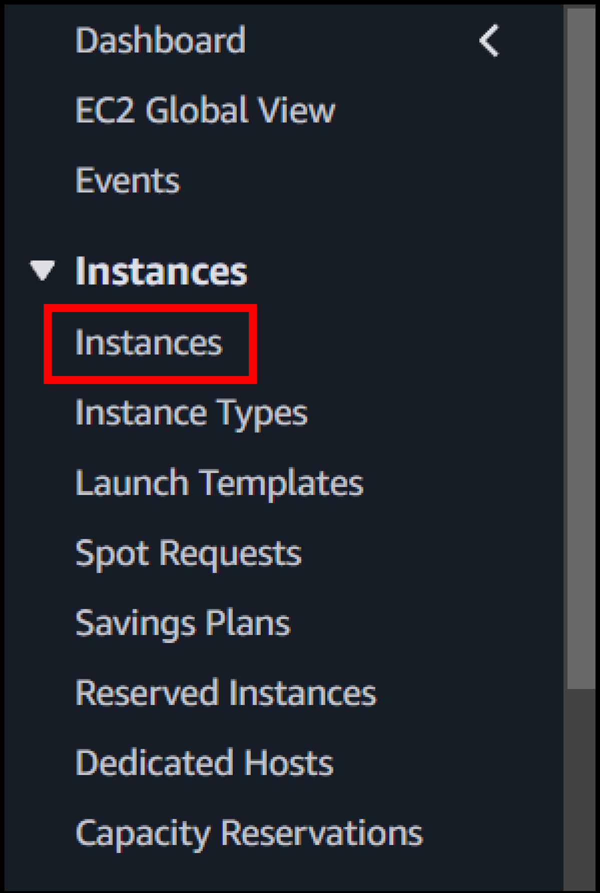


13. Scroll down and click on "Create Auto Scaling Group".

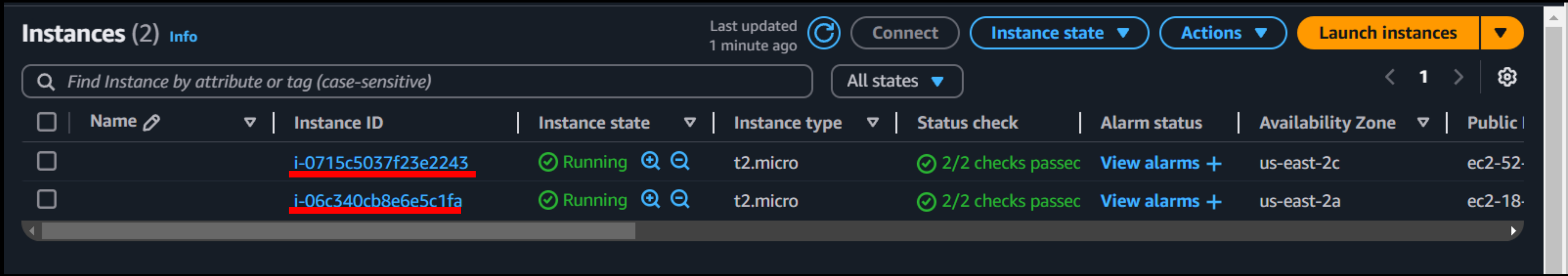


IV. VERIFYING AUTO SCALING GROUP ✨

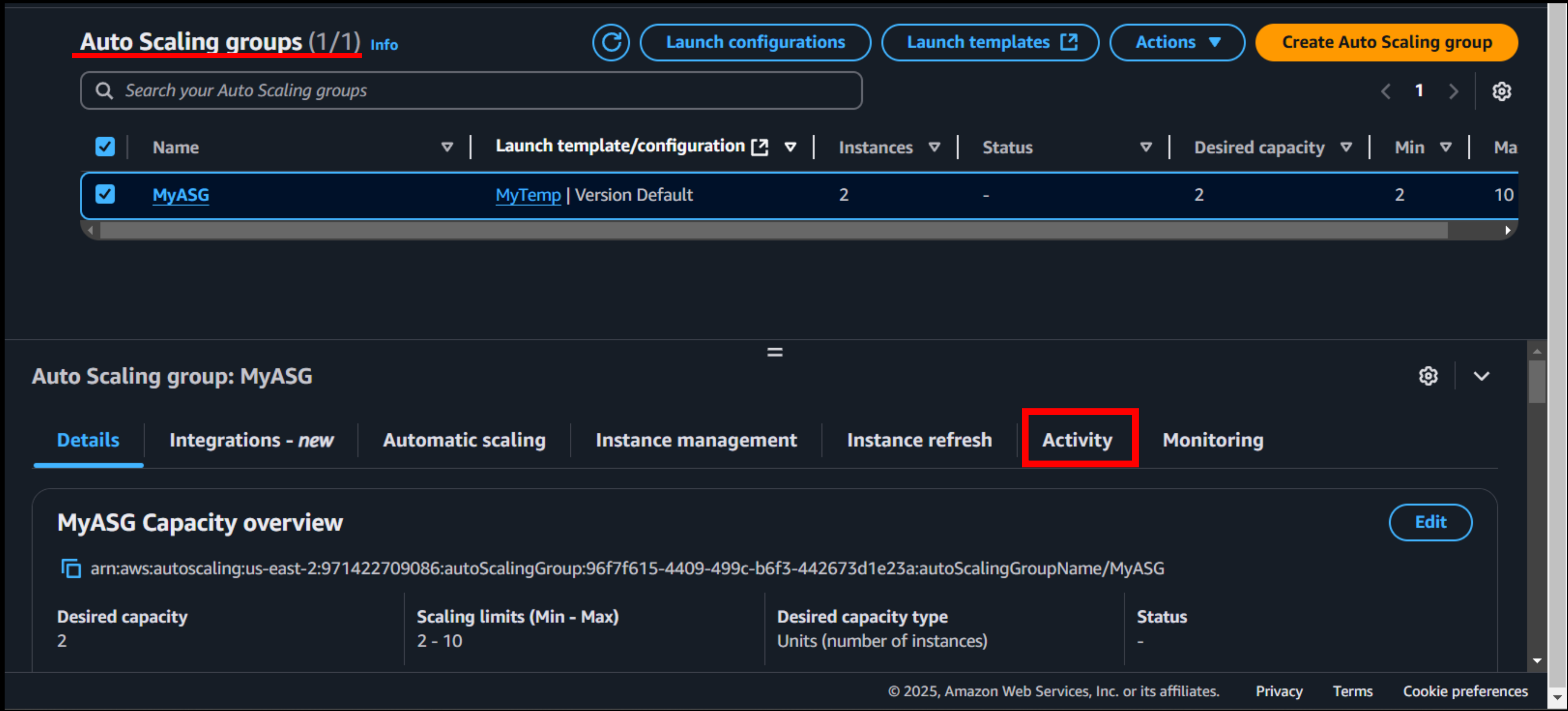
1. Navigate to the EC2 Instance section.



2. Verify that the instances are being created.



3. Go back to the ASG Dashboard.



4. Navigate to "Activity" and compare the Instance IDs.

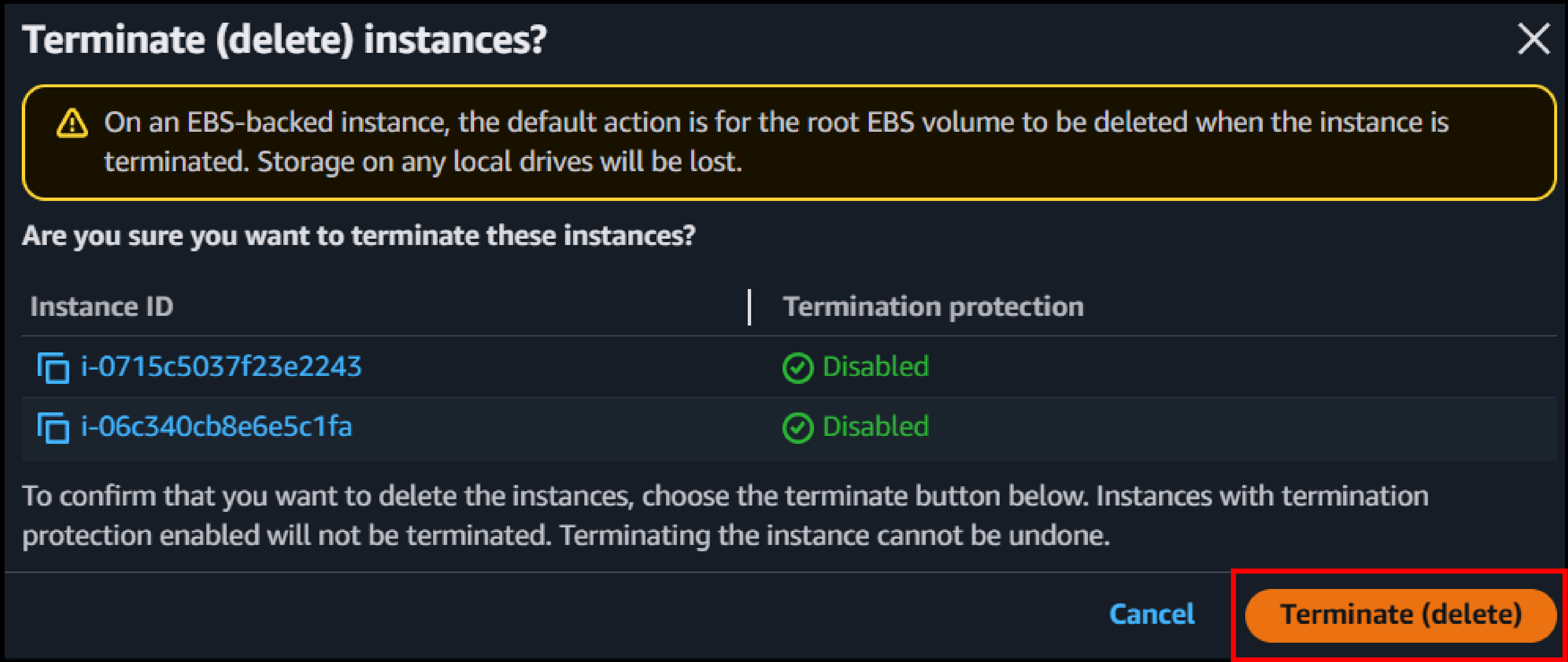
Status	Description	Cause	Start time	En
Successful	Launching a new EC2 instance: i-0715c5037f23e2243	At 2025-02-02T14:39:47Z a user request created an AutoScalingGroup changing the desired capacity from 0 to 2. At 2025-02-02T14:39:49Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 0 to 2.	2025 February 02, 08:09:51 PM +05:30	20 Fe 08 +0
Successful	Launching a new EC2 instance: i-06c340cb8e6e5c1fa	At 2025-02-02T14:39:47Z a user request created an AutoScalingGroup changing the desired capacity from 0 to 2. At 2025-02-02T14:39:49Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 0 to 2.	2025 February 02, 08:09:51 PM +05:30	20 Fe 08 +0

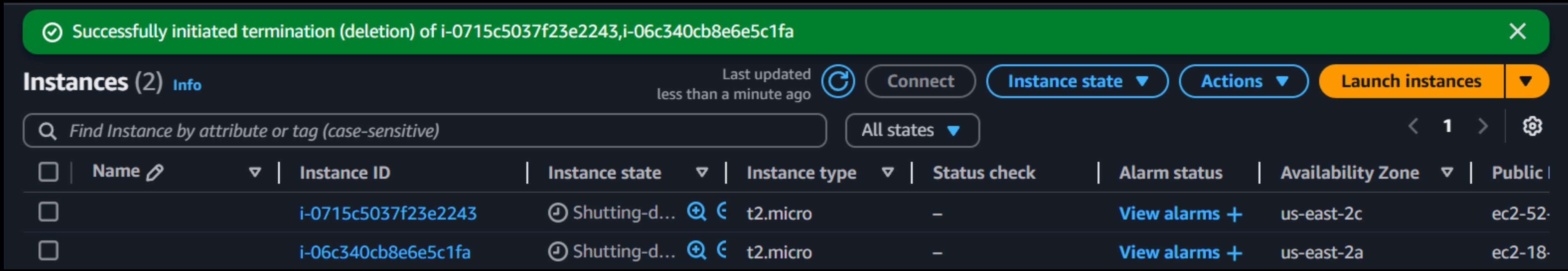
Instance ID
i-0715c5037f23e2243
i-06c340cb8e6e5c1fa

Description
Launching a new EC2 instance: i-0715c5037f23e2243
Launching a new EC2 instance: i-06c340cb8e6e5c1fa

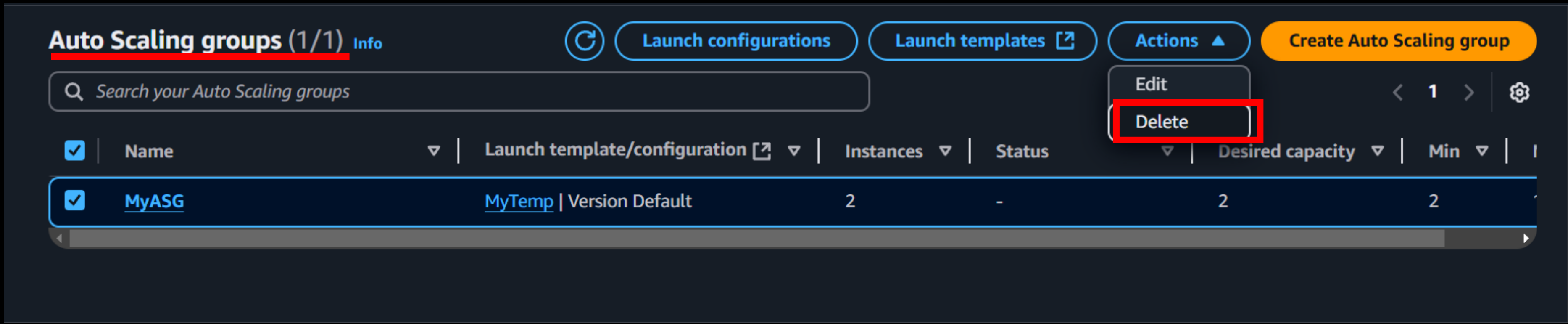
V. CLEANING UP RESOURCES ♻️

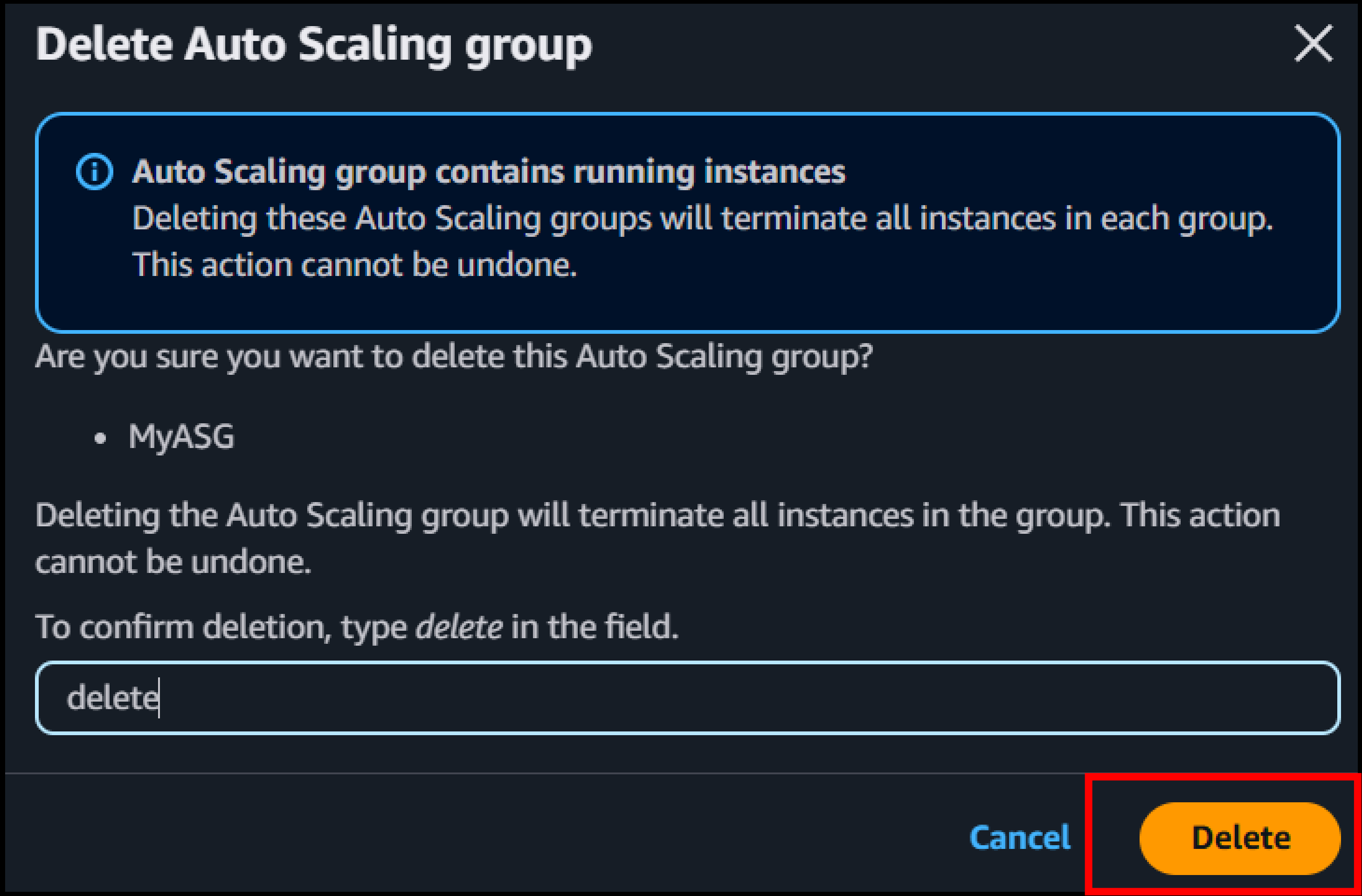
1. Terminate the instances.

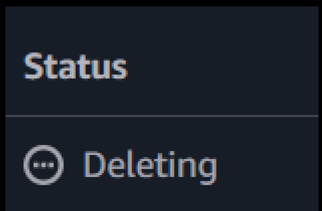




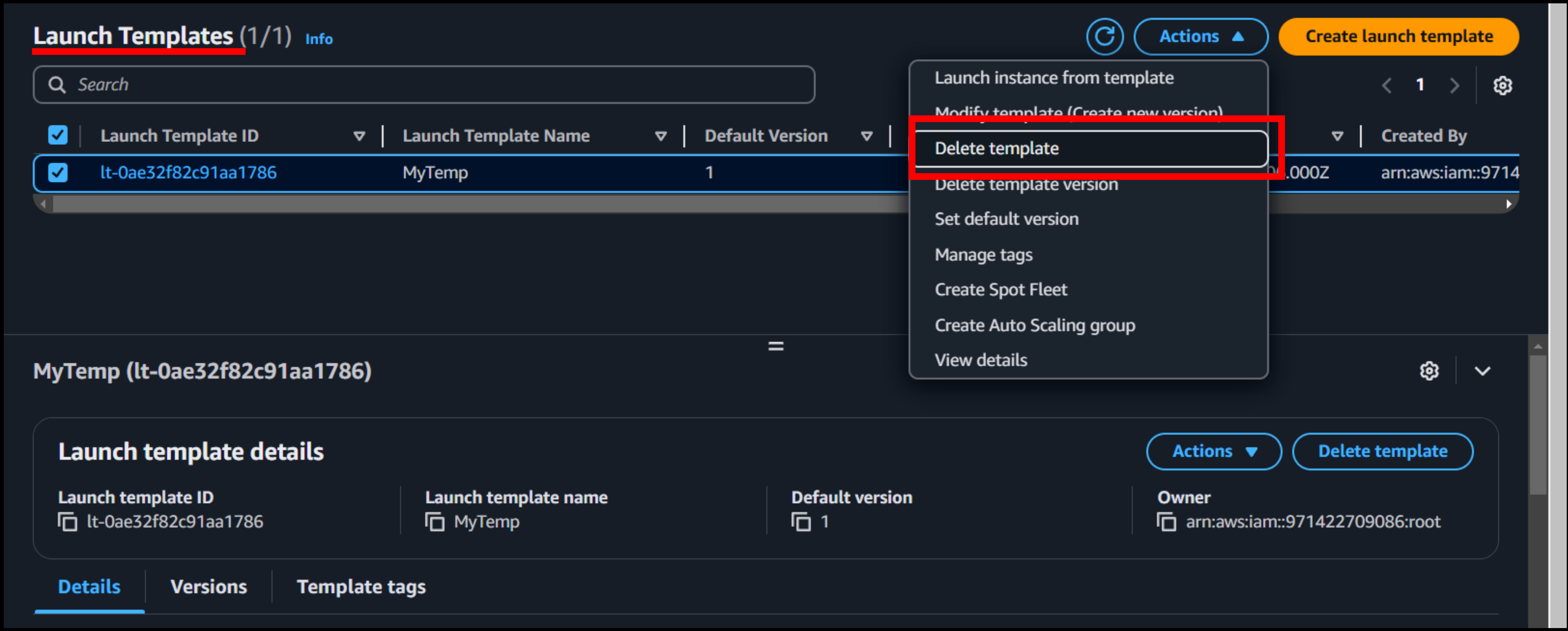
2. Delete the Auto Scaling Group (ASG).

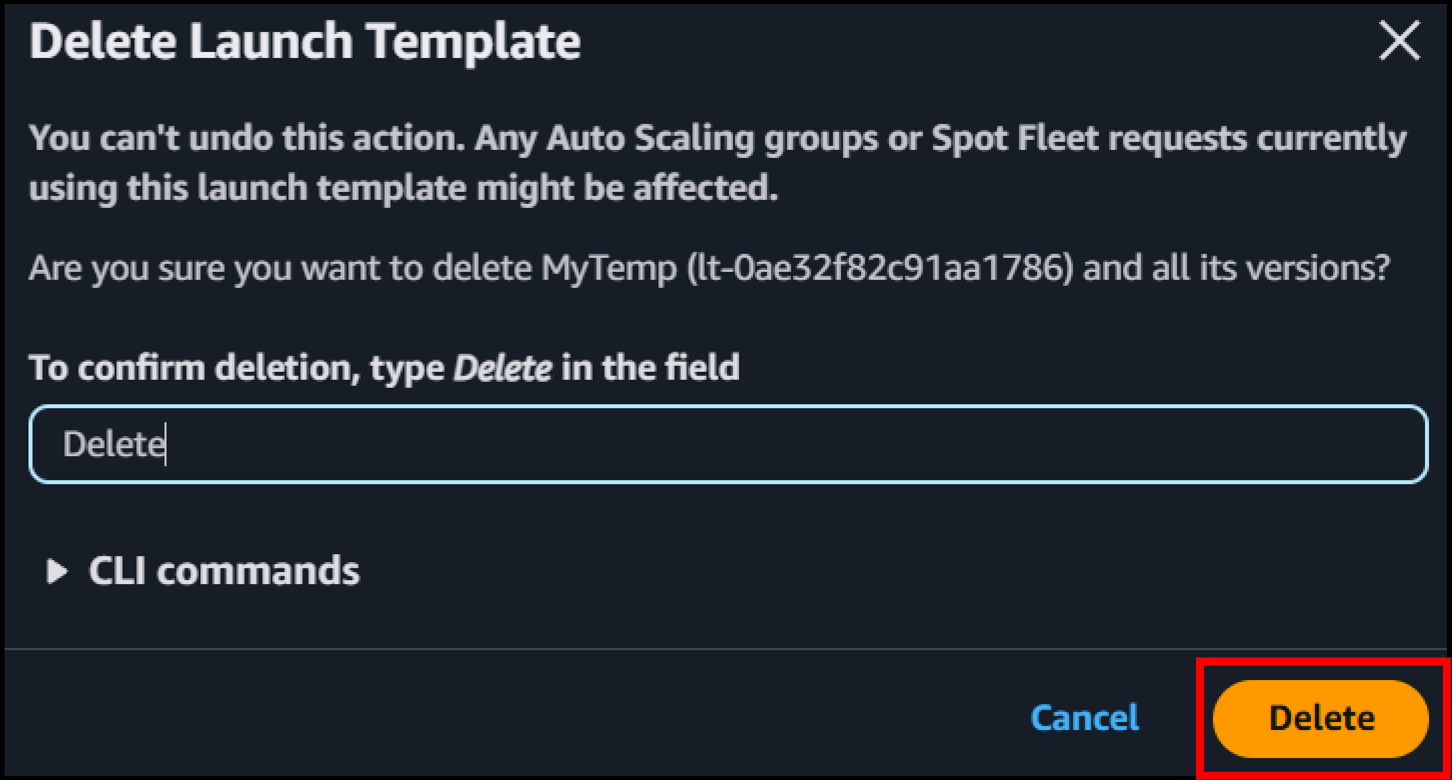


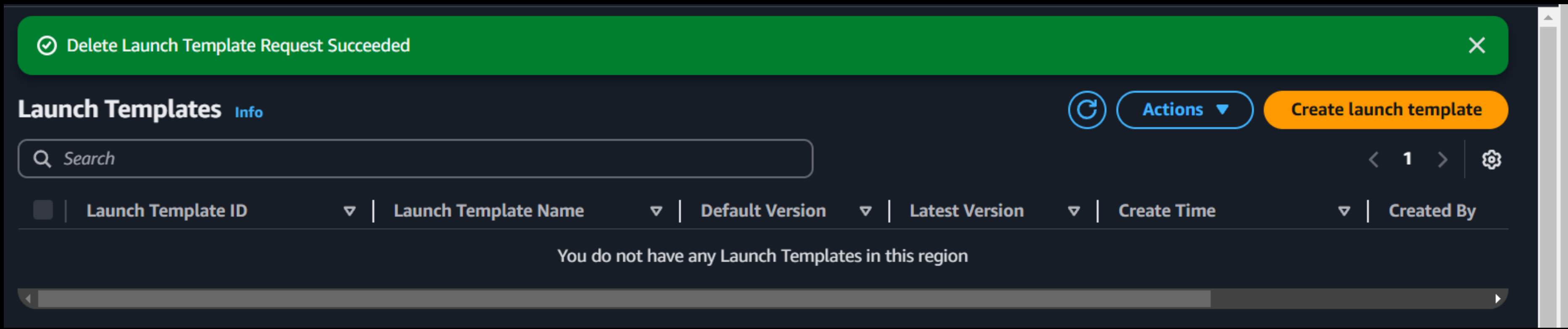

Delete Auto Scaling group
Auto Scaling group contains running instances
Deleting these Auto Scaling groups will terminate all instances in each group. This action cannot be undone.
Are you sure you want to delete this Auto Scaling group?
MyASG
Deleting the Auto Scaling group will terminate all instances in the group. This action cannot be undone.
To confirm deletion, type delete in the field.
delete
Cancel
Delete


Status
Deleting

3. Delete the Launch Template.




Delete Launch Template
You can't undo this action. Any Auto Scaling groups or Spot Fleet requests currently using this launch template might be affected.
Are you sure you want to delete MyTemp (lt-0ae32f82c91aa1786) and all its versions?
To confirm deletion, type Delete in the field
Delete
CLI commands
Cancel
Delete


Delete Launch Template Request Succeeded
Launch Templates
Info
Actions
Create launch template
Search
1
Launch Template ID
Launch Template Name
Default Version
Latest Version
Create Time
Created By
You do not have any Launch Templates in this region

CONCLUSION 🎯

Congratulations 🎉 You have successfully created an Auto Scaling Group (ASG) in AWS, ensuring that your infrastructure can dynamically scale based on demand. This enhances reliability, optimizes costs, and improves overall efficiency. Keep exploring AWS to refine your auto-scaling strategies 🚀

Efficiency is doing things right; effectiveness is doing the right things 💡

– Peter Drucker